



$\therefore$ 最大流与最小截量均为 14.

2. (1) $A = \begin{pmatrix} 1 & 0 & 3 & 4 \\ -1 & 4 & 0 & 1 \\ 2 & 2 & 2 & 3 \\ 0 & 4 & 1 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 & 0 \\ -1 & 4 \\ 2 & 2 \\ 0 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 2 & 2 \\ 0 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 2 \\ 0 \end{pmatrix} \Rightarrow (2)$

$\therefore$ 矩阵对策的解为 (x_3, β_1)

(2) $A = \begin{pmatrix} 3 & 4 & 0 & 3 & 0 \\ 5 & 0 & 2 & 5 & 9 \\ 7 & 3 & 9 & 5 & 9 \\ 4 & 6 & 8 & 7 & 6 \\ 6 & 0 & 8 & 8 & 3 \end{pmatrix} \Rightarrow \begin{pmatrix} 7 & 3 & 9 & 5 & 9 \\ 4 & 6 & 8 & 7 & 6 \\ 6 & 0 & 8 & 8 & 3 \end{pmatrix} \Rightarrow \begin{pmatrix} 7 & 3 \\ 4 & 6 \\ 6 & 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 7 & 3 \\ 4 & 6 \end{pmatrix}$

$$\therefore \begin{cases} 7x_1 + 4x_2 = V \\ 3x_1 + 6x_2 = V \\ x_1 + x_2 = 1 \\ x_1, x_2 \geq 0 \end{cases} \quad \begin{cases} 7y_1 + 3y_2 = V \\ 4y_1 + 6y_2 = V \\ y_1 + y_2 = 1 \\ y_1, y_2 \geq 0 \end{cases}$$

$\therefore x_1^* = \frac{1}{3}, x_2^* = \frac{2}{3}, y_1^* = \frac{1}{2}, y_2^* = \frac{1}{2}, V = 5$

$\therefore$ 解为 $X^* = (0, 0, \frac{1}{3}, \frac{2}{3}, 0)^T, Y^* = (\frac{1}{2}, \frac{1}{2}, 0, 0, 0)^T, V = 5.$